

Ongoing Work on Contextual Neural Machine Translation

Duc Thanh Tran

`tran@i6.informatik.rwth-aachen.de`

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**Human Language Technology and Pattern Recognition
Lehrstuhl für Informatik 6
Computer Science Department
RWTH Aachen University, Germany**

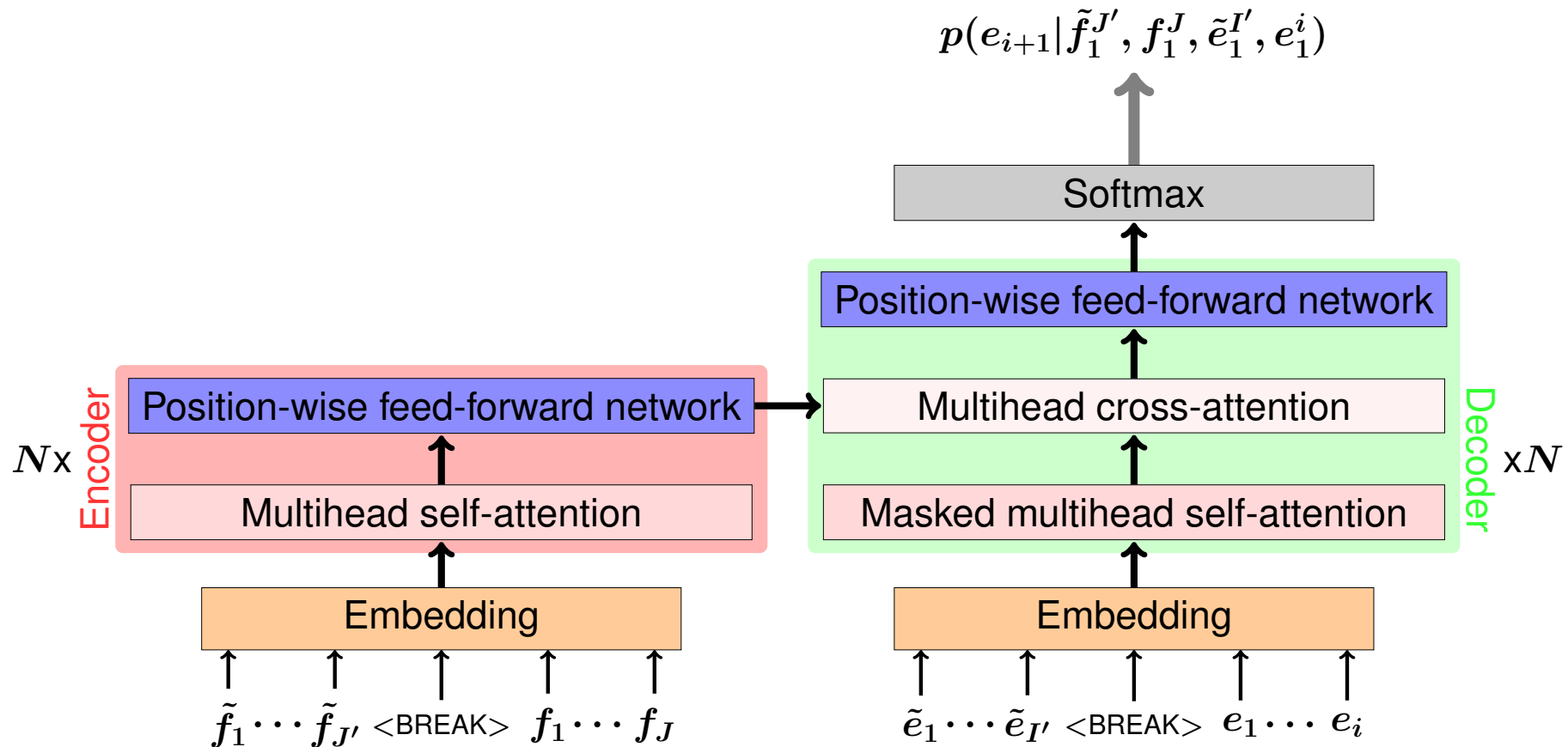
- ▶ Most neural machine translation (NMT) systems translate sentence by sentence
- ▶ Make use of surrounding sentences to exploit context information
 - ▷ reduce ambiguities in translation choices
 - ▷ example: cross-sentence anaphora in English → German

Source sentences	Correct translation of 'it'
<i>The monkey ate the banana. Because it was hungry.</i>	er
<i>The monkey ate the banana. Because it ripe.</i>	sie
<i>The monkey ate the banana. Because it was tea-time.</i>	es

- ▶ **Single encoder: concatenate sentences on source-/target side**
[Tiedemann & Scherrer 17, Agrawal & Turchi⁺ 18]
- ▶ **Multi-Encoder with attention mechanism**
 - ▷ **Feed surrounding sentences into separate encoders and combine context vectors:**
 - **concatenation (like in [Zoph & Knight 16])**
 - **gated sum [Bawden & Sennrich⁺ 18]**
 - **hierarchical attention [Bawden & Sennrich⁺ 18]**

Remark: above methods only consider *local* context

Transformer Architecture with Concatenated Context Sentences



- ▶ one previous context-sentence on the source ($\tilde{f}_1^{J'}$) and target ($\tilde{e}_1^{I'}$) side
- ▶ **<BREAK>** token separates the sentences

► [Build multi-encoder]

- one previous context-sentence on the source ($\tilde{f}_1^{J'}$) is encoded separately

1. Reproduce results from [Agrawal & Turchi⁺ 18]

- ▶ Transformer model
- ▶ IWSLT 2017 English→Italian
- ▶ concatenate sentences with special token **<BREAK>**,
example: sentence_1 **<BREAK>** sentence_2

2. Compress context-sentences by pruning

- ▶ first n most-common words in training corpus
- ▶ pre-defined stopword list

3. Test method of concatenation (+ pruning) on WMT 2018 German→English data as well

Concatenation

(a) Single target sentence.

Source	→	Target
1 previous + current	→	current
2 previous + current	→	current
1 previous + current + 1 next	→	current
current	→	current

(b) Multiple target sentences.

Source	→	Target
1 previous + current	→	1 previous + current
1 previous + current + 1 next	→	1 previous + current

Dropping Words in Context Sentences

- ▶ Remove punctuations first
- ▶ Drop $MC \in \mathbb{N}$ most common words from the training corpus
 - ▷ $DW \equiv$ Percentage of dropped words in a corpus
 - ▷ For $MC = 90$ we obtain $DW = 57\%$ on IWSLT En $\rightarrow$ It data

And that led us directly to the Olympic Village .



led directly Olympic Village

- ▶ Concatenate shortened context-sentences to current one that is untouched

Data Statistics

		IWSLT'17 EN-IT		WMT'18 DE-EN	
		EN	IT	DE	EN
Train.	Sentences	232K		5.9M	
	Running words	4.7M	4.4M	160.9M	157.7M
Dev.	Sentences	<i>tst2010</i> 1.5K		<i>newstest2016</i> 3K	
	Running words	31K	29K	77K	73K
Test	Sentences	<i>tst2017</i> 1.1K		<i>newstest2017</i> 3K	
	Running words	22K	20K	76K	73K

- ▶ **Transformer model [Vaswani & Shazeer⁺ 17] with 6 encoder/decoder layers**
- ▶ **Model dimensionality: 512**
- ▶ **Feed-forward network dimensionality: 2048**
- ▶ **Attention-heads: 8**

- ▶ **Adam optimizer [Kingma & Ba 14]**
 - ▷ **learning rate: 0.0002**
 - ▷ **2,000 warmup steps**

- ▶ **32,000 Byte-pair encoding merges [Sennrich & Haddow⁺ 16]**

Results: IWSLT 2017 English→Italian

Concatenation baselines.

			<i>tst2017</i>	
			BLEU [%]	TER [%]
current	→	current	29.89	52.0
1 previous + current	→	current	30.02	51.8
2 previous + current	→	current	30.00	52.1
1 previous + current + 1 next	→	current	30.46	52.0
current + 1 next	→	current	29.93	51.9
1 previous + current	→	1 previous + current	30.64	51.0
1 previous + current + 1 next	→	1 previous + current	30.64	51.4

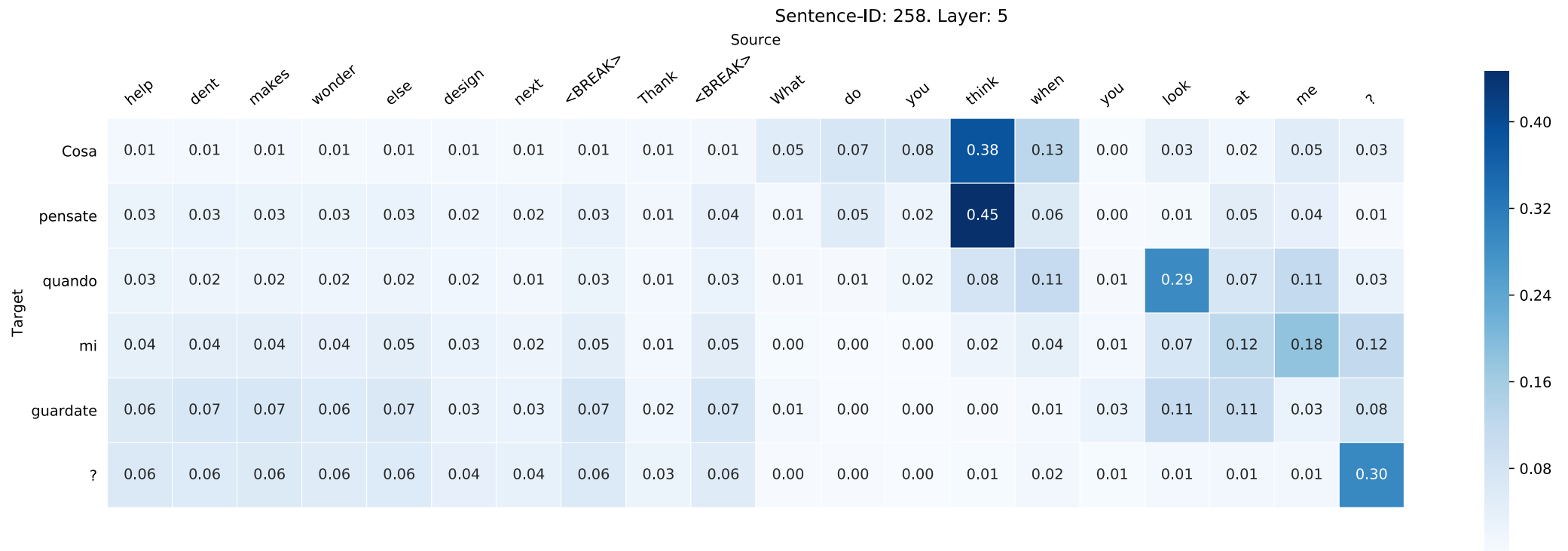
Dropping most common words experiments: *single* target sentence.

Source	→ Target			<i>tst2017</i>	
		<i>MC</i>	<i>DW</i>	BLEU [%]	TER [%]
current	→ current	0	0%	29.89	52.0
1 previous + current	→ current	0	0%	30.02	51.8
		100	59%	30.03	52.0
2 previous + current	→ current	0	0%	30.00	52.1
		90	57%	30.35	51.4
1 previous + current + 1 next → current		0	0%	30.46	52.0
		90	57%	30.14	51.8
current + 1 next → current		0	0%	29.93	51.9
		100	59%	30.00	51.6

Dropping most common words experiments: *multiple* target sentences.

Source	→	Target	<i>tst2017</i>			
			<i>MC</i>	<i>DW</i>	BLEU [%]	TER [%]
1 previous + current	→	1 previous + current	0	0%	30.64	51.0
			90	57%	30.09	52.0
1 previous + current + 1 next	→	1 previous + current	0	0%	30.64	51.4
			150	64%	30.18	51.6

IWSLT English→Italian: Cross-Attention Examples



Sentence-ID: 238. Layer: 3



Results: WMT 2018 German→English

Concatenation baselines.

			<i>newstest2017</i>	
			BLEU [%]	TER [%]
current	→	current	30.36	51.0
1 previous + current	→	current	30.13	51.4
2 previous + current	→	current	30.32	51.2
1 previous + current + 1 next	→	current	30.28	51.2
current + 1 next	→	current	30.15	51.0
1 previous + current	→	1 previous + current	30.10	51.2
1 previous + current + 1 next	→	1 previous + current	30.18	51.2

Dropping most common words experiments: *single* target sentence.

Source	→ Target			<i>newstest2017</i>	
		<i>MC</i>	<i>DW</i>	BLEU [%]	TER [%]
current	→ current	0	0%	30.36	51.0
1 previous + current	→ current	0	0%	30.13	51.4
		300	57%	29.98	50.86
2 previous + current	→ current	0	0%	30.32	51.2
		300	57%	27.66	60.08
1 previous + current + 1 next → current		0	0%	30.28	51.2
		300	57%	30.06	51.8
current + 1 next → current		0	0%	30.15	51.0
		300	57%	30.20	51.2

Dropping most common words experiments: *multiple* target sentences.

Source	→	Target			<i>newstest2017</i>	
			<i>MC</i>	<i>DW</i>	BLEU [%]	TER [%]
1 previous + current	→	1 previous + current	0	0%	30.10	51.0
			300	57%	????	???
1 previous + current + 1 next	→	1 previous + current	0	0%	30.18	51.4
			300	57%	27.31	54.4

- ▶ **Concatenation of sentences yield slight performance increases with regards to BLEU and TER metrics**
 - ▷ IWSLT En→It: up to +0.7 BLEU improvement
 - ▷ WMT De→En: no improvement
- ▶ **Simple compression by dropping words according to frequencies**
 - ▷ IWSLT En→It: up to +1 BLEU improvement
 - ▷ WMT De→En: no improvement
- ▶ **English language is easier in terms of linguistic properties**
- ▶ **Dropping only stopwords does not yield improvements**

- ▶ **Implementation of multi-encoder approaches**
[Bawden & Sennrich⁺ 18, Voita & Sennrich⁺ 18]
- ▶ **Consider global context information: e.g. topic embedding**
- ▶ **Test approaches in discourse-heavy data sets**
like OpenSubtitles [Lison & Tiedemann 16]
- ▶ **More sophisticated evaluation with regards to discourse-phenomena**
 - ▷ pronoun translation
 - ▷ co-reference

Thank you for your attention

Duc Thanh Tran

`tran@i6.informatik.rwth-aachen.de`

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Appendix: Results from [Agrawal & Turchi⁺ 18]

► [blub]

Appendix: IWSLT English→Italian

Dropping Stopwords

► [blub]